

Plant Anatomy

1. Introduction

Plant anatomy is the branch of botany that studies the internal structure of plants, including cells, tissues, organs, and their arrangement. It explains how roots, stems, leaves, flowers, and woody parts are built and how their structures support function.

In BSc Botany, plant anatomy is a core subject because it connects morphology, physiology, development, and taxonomy. It helps students understand how internal structures adapt to water conduction, food transport, support, protection, and secondary growth.

2. Scope and importance

Plant anatomy is important in agriculture, forestry, taxonomy, pharmacognosy, ecology, and plant breeding. Anatomical features help identify plant groups, understand adaptations, and interpret evolutionary relationships.

It is also essential for explaining how xylem and phloem work, how cambium produces secondary tissues, and how protective layers are formed in woody plants.

3. Plant tissues

Plant tissues are groups of cells with common origin and similar function. They are broadly divided into meristematic and permanent tissues.

3.1 Meristematic tissue

Meristematic cells divide actively and help in growth. They are small, thin-walled, with dense cytoplasm and prominent nuclei.

Types of meristem

- Apical meristem.
- Intercalary meristem.
- Lateral meristem.

3.2 Permanent tissue

Permanent tissues are derived from meristems and have lost the capacity for division. They may be simple or complex.

Simple permanent tissues

- Parenchyma.
- Collenchyma.
- Sclerenchyma.

Complex permanent tissues

- Xylem.
- Phloem.

4. Epidermal tissue system

The epidermis is the outer protective layer of young plant organs. It consists of epidermal cells, stomata, and trichomes.

Functions

- Protection.
- Prevention of water loss.
- Gas exchange through stomata.
- Absorption in root hairs.
- Defense through trichomes.

5. Ground tissue system

Ground tissue lies between epidermis and vascular tissues. It includes cortex, endodermis, pericycle, pith, and medullary rays depending on the organ.

Functions

- Storage.
- Photosynthesis.
- Support.
- Lateral transport.
- Mechanical strength.

6. Vascular tissue system

The vascular system consists of xylem and phloem. It is the transport system of the plant body and is especially well developed in vascular plants.

Xylem

Xylem conducts water and minerals and also provides mechanical support. It consists of tracheids, vessels, xylem parenchyma, and xylem fibres.

Phloem

Phloem transports food materials, especially sugars, from source to sink regions. It consists of sieve tubes, companion cells, phloem parenchyma, and phloem fibres.

7. Xylem in detail

Xylem is a complex tissue responsible for conduction of water and minerals from roots to aerial parts. It also helps in support due to thick lignified walls.

Elements

- Tracheids.
- Vessels.
- Xylem parenchyma.
- Xylem fibres.

Types

- Protoxylem.
- Metaxylem.

8. Phloem in detail

Phloem is the food-conducting tissue of plants. It translocates sucrose and other organic solutes from leaves to growing and storage regions.

Elements

- Sieve elements.
- Companion cells.
- Phloem parenchyma.
- Phloem fibres.

9. Tissue organization in root

The root is specialized for absorption, anchorage, and storage. It has root cap, epidermis, cortex, endodermis, pericycle, and vascular cylinder.

Dicot root

- Star-shaped xylem.
- Phloem between xylem arms.
- Cambium appears during secondary growth.

Monocot root

- Many xylem bundles.
- Large pith.
- Usually no secondary growth.

10. Tissue organization in stem

The stem supports leaves, flowers, and fruits and conducts materials. It usually shows epidermis, cortex, endodermis, pericycle, vascular bundles, and pith.

Dicot stem

- Vascular bundles arranged in a ring.
- Cambium present in bundles.
- Secondary growth common.

Monocot stem

- Vascular bundles scattered.
- Cambium absent.
- Secondary growth usually absent.

11. Tissue organization in leaf

Leaves are the main photosynthetic organs. A typical leaf shows upper epidermis, mesophyll, vascular bundles, and lower epidermis.

Mesophyll

- Palisade parenchyma.
- Spongy parenchyma.

Venation

- Reticulate venation in dicots.
- Parallel venation in monocots.

12. Secondary growth

Secondary growth is the increase in girth of roots and stems due to the activity of lateral meristems. It is typical of dicots and gymnosperms.

Main lateral meristems

- Vascular cambium.
- Cork cambium.

13. Vascular cambium

Vascular cambium is the meristem responsible for producing secondary xylem inward and secondary phloem outward. It forms a cambial ring in dicot stems and roots.

Formation of cambial ring

- Fascicular cambium develops within vascular bundles.
- Interfascicular cambium develops between bundles.
- These join to form a continuous ring.

Activity

- More secondary xylem is produced than secondary phloem.
- This causes wood formation.

14. Secondary xylem and wood

Secondary xylem forms the bulk of wood. It includes vessels, tracheids, fibres, and parenchyma, and is responsible for conduction and support.

Features

- Annual rings.
- Heartwood and sapwood.
- Wood rays for radial transport.
- Mechanical strength.

15. Cork cambium and periderm

Cork cambium, or phellogen, develops in outer tissues when the stem increases in girth. It produces cork (phellem) outward and phelloderm inward. The three tissues together form the periderm.

Significance

- Replaces epidermis in woody plants.
- Prevents water loss.
- Protects against pathogens and injury.

16. Secondary growth in root

In dicot roots, secondary growth begins with the formation of vascular cambium from tissues near the xylem and pericycle. This cambium forms secondary xylem inward and secondary phloem outward.

17. Anatomy of dicot and monocot organs

Dicot root

- Central xylem core.
- Few xylem bundles.
- Secondary growth present.

Monocot root

- Many xylem bundles.
- Large central pith.
- No typical secondary growth.

Dicot stem

- Ring of vascular bundles.
- Cambium present.
- Secondary growth common.

Monocot stem

- Scattered vascular bundles.
- Cambium absent.
- No normal secondary growth.

18. Adaptive anatomy

Anatomy also reflects environmental adaptation. Xerophytes often show thick cuticle, sunken stomata, and mechanical tissue, while hydrophytes show reduced supporting tissues and large air spaces.

19. Summary

Plant anatomy explains the internal organization of plant bodies from tissues to organs and is essential for understanding transport, support, protection, and growth. Secondary growth, vascular cambium, and cork cambium are especially important in advanced botany because they explain how woody plants increase in girth and develop protective tissues.